

Data management maturity assessments

DATA MANAGEMENT CONCEPTS



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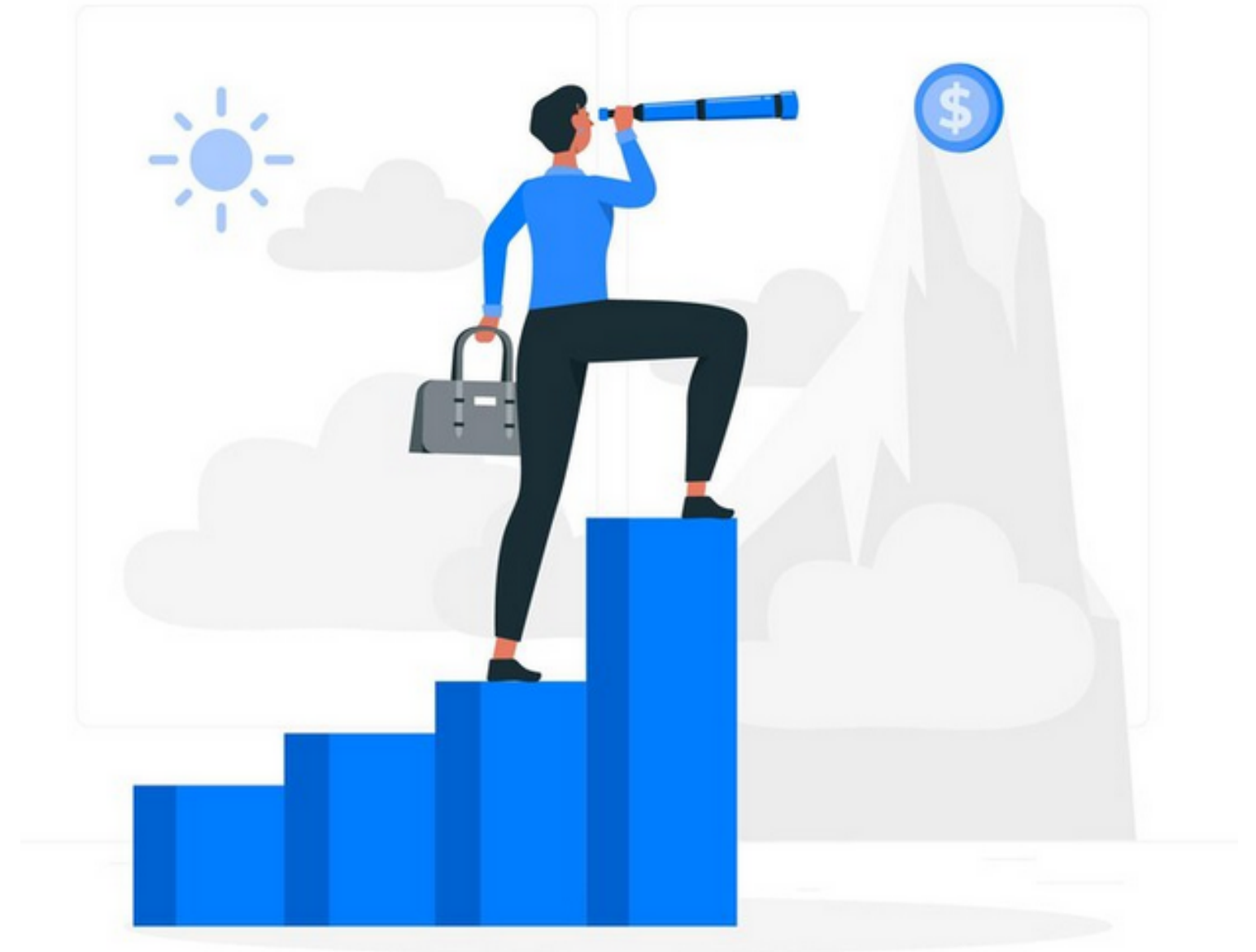
Data maturity versus data management maturity

Data maturity

- How well an organization understands, values, and uses its data
- Looks at the **big picture** of data capabilities

Data management maturity

- Effectiveness of data management practices i.e. **processes, policies, and technologies used**
- **Thorough analysis** of data practices and strategies



¹ Images in this lesson were created using Freepik

Why assess data management maturity?

- Blueprint for optimizing data management practices
- Implement standardized assessment procedures
- Frameworks:
 - **Capability Maturity Model Institute - Data Management Maturity Model (CMMI-DMM)**
 - **Data Management Capability Assessment Model EDM Council (DCAM)**



CMMI - Data Management Maturity Model

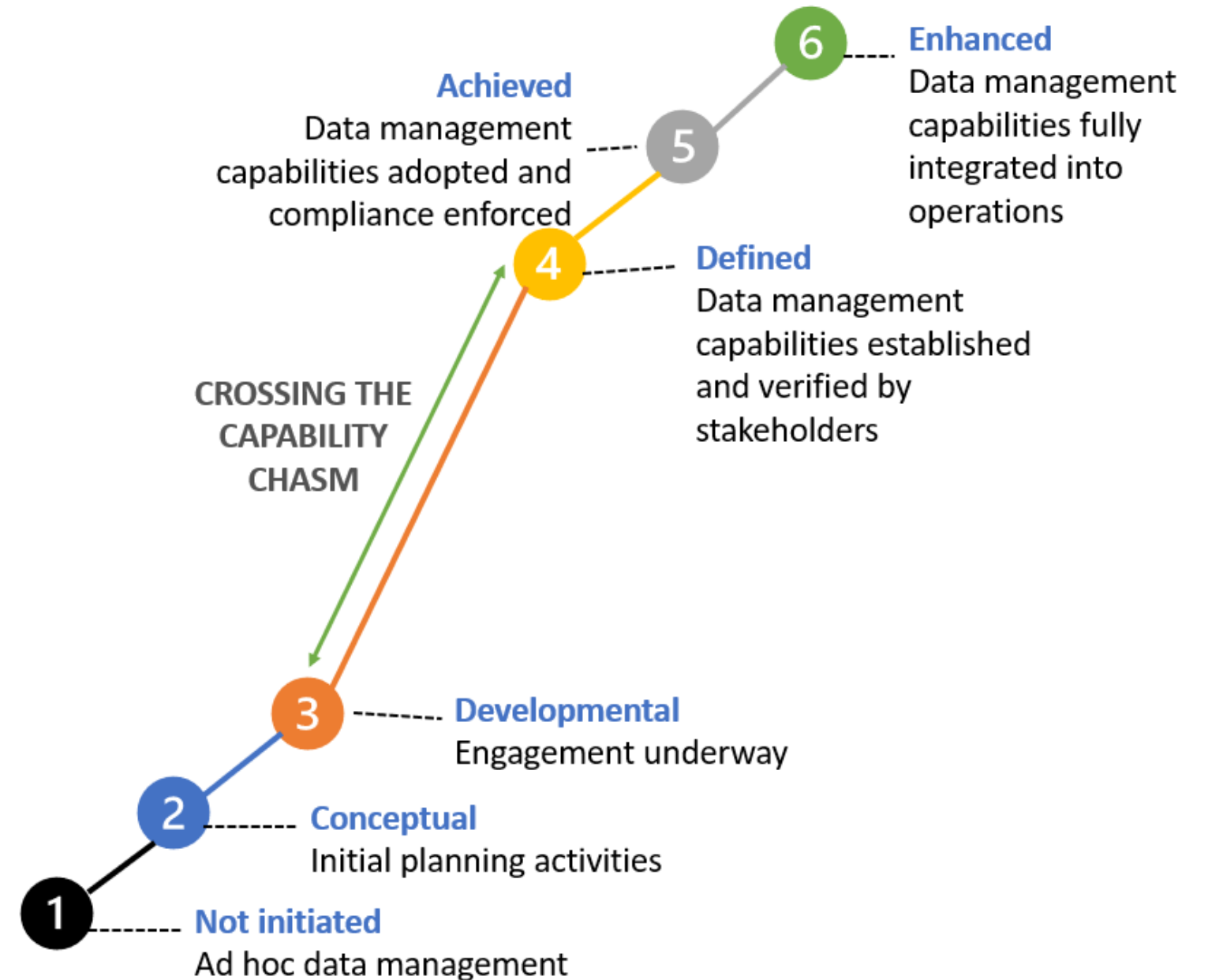
- Structured framework for improving data capabilities
- **5 maturity levels** - guides organizations from **initial** practices to **optimized**, well-defined processes
- Covers **25** process areas, including quality, integration, architecture among others



¹ cmmiinstitute.com

EDM Council (DCAM)

- Focuses on data governance, quality, life cycle, architecture, metadata management and more
- Scoring system progressing from **Not initiated** to **Enhanced**
- Balanced approach: emphasizes interplay of **people, processes, and technology**



¹ edmcouncil.org

Tailored approaches to assessment

- Customized to specific organizational needs
- **Self-evaluations** - internal assessments using predefined frameworks
- **Third-party assessments** - objective external evaluation
- **Benchmarking** - comparative analyses against industry



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Data management operating models

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Unpacking centralized models

- Tasks **centralized** under a specific authority or team
- Ensures **uniformity** and **control**
- **Eliminates redundancy** across departments
- **Provides clear guidelines** for data handling
- Possible limitations include **bureaucracy** and **inflexibility** adapting to specific needs
- Best suited to organizations prioritizing **standardization, compliance, and unified approach**



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Understanding decentralized models

- Dispersed data responsibilities
- Independence and local decision-making
- **Flexible** response to departmental needs and instills a **sense of ownership**
- Challenges include **consistency hurdles, standardization issues**
- **Communication and alignment** are paramount
- Suitable for organizations with **varied units** and **distinct data requirements**



Navigating hybrid models

- **Blended** approach combining elements from both **centralized** and **decentralized** models
- Aims to find equilibrium between **flexibility**, **autonomy**, **control**, and **standardization**
- May have central management for overall governance and strategic decisions
- Various units may have autonomy over specific data tasks
- Beneficial for organizations requiring both **central oversight** and **decentralized flexibility**



Considerations for implementation

1. Organizational culture
2. Scalability
3. Data sensitivity and security
4. Flexibility and responsiveness
5. Technology infrastructure compatibility
6. Integration requirements
7. Regulatory compliance
8. Cost implications



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Ethical data handling

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The significance of consent



- **Consent** is a freely given agreement to something
- Utilizing **explicit opt-in mechanisms** ensures active agreement
- **Avoid assumptions** through pre-selected options
- Individuals should be able to **withdraw agreement** without adverse consequences
- Maintaining **detailed consent records** is crucial

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Promoting awareness in data handling

- Awareness promotion is a key ethical practice
- Involves **clear communication** to ensure individuals understand data usage
- Training programs and initiatives **enhance data literacy**
- Enable concerns and clarification through **accessible feedback channels**



Ensuring data privacy



- Protecting personal information at every stage of its journey
- Adhering to established **regulations**
- Gathering only the information strictly **necessary** to achieve specific purposes
- **Informing individuals** about data practices and any changes implemented
- Building privacy considerations into the core of technology systems

Upholding individual data access rights

- Individuals have the right to access their **personal data**
- Response to access requests must be **prompt**
- **Secure access channels** are essential
- Procedures should be **user-friendly** and **fees minimized**
- Explicit **explanations** are required for access denials
- **Identity verification** processes are crucial



Accountability in data management



- **Designated data stewards** should oversee ethical data use and protection
- **Audit trails** tracking data handling activities
- **Detailed documentation** provide evidence of ethical compliance
- Accountability extends to **third-party data processors**
- **Regular reviews** drive continuous improvement

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The future of data management

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What does the future hold?

- Real-time data management
- Cloud-based data management
- Data mesh



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Real-time data management



- Real-time data management for swift decision-making
- Social media: Instant personalized content and ads
- Stock exchanges: Real-time market trend analysis
- Powerful and accessible data processing tools

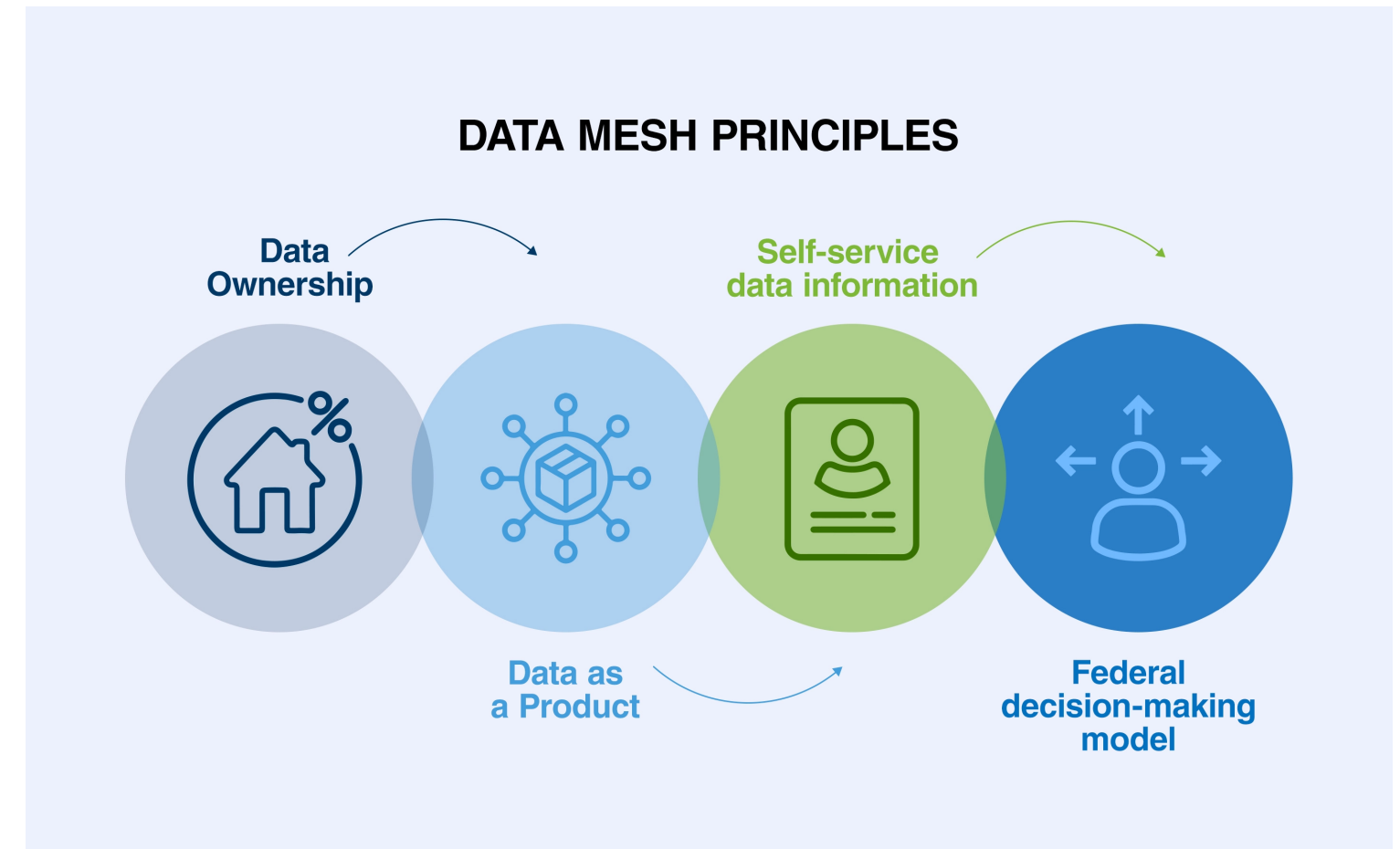
Cloud-based data management

- Redefining data storage and analysis
- Greater integration, security, and user-friendly interfaces
- Enhanced scalability, cost-efficiency, and accessibility
- Level playing field for organizations of all sizes



Data mesh architecture

- Introduced in 2019 by Zhamak Dehghani
- Data ownership, data as a product, self-service data information, federal decision-making model
- Enhanced quality, accessibility and security
- More agile and effective
- Improved scalability, responsiveness, and data innovation



Preparing for the future



- Cultivate a data-driven culture
- Continuous learning
- Robust data security
- Collaboration is key
- Stay adaptable

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Recap

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Chapter 1: Unveiling data management

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- Data management definition
 - Benefits of data management
 - Data life cycle stages
 - Key principles for effective data management

Chapter 2: Pillars of data management

Data management pillars
e.g. data quality, metadata
management, data
security and privacy, data
modeling

Aiken's pyramid and the
different phases

Chapter 3: Mastering maturity, embracing ethics, and shaping your future

- Data management maturity assessments e.g. CMMI- DMM , EDM Council DCAM
- Data management operating models e.g. decentralized, centralized, hybrid
- Ethical data handling
- Future of data management

What's next?

Courses in data literacy

- Data governance concepts
- Introduction to data quality
- Introduction to data privacy

Congratulations!

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